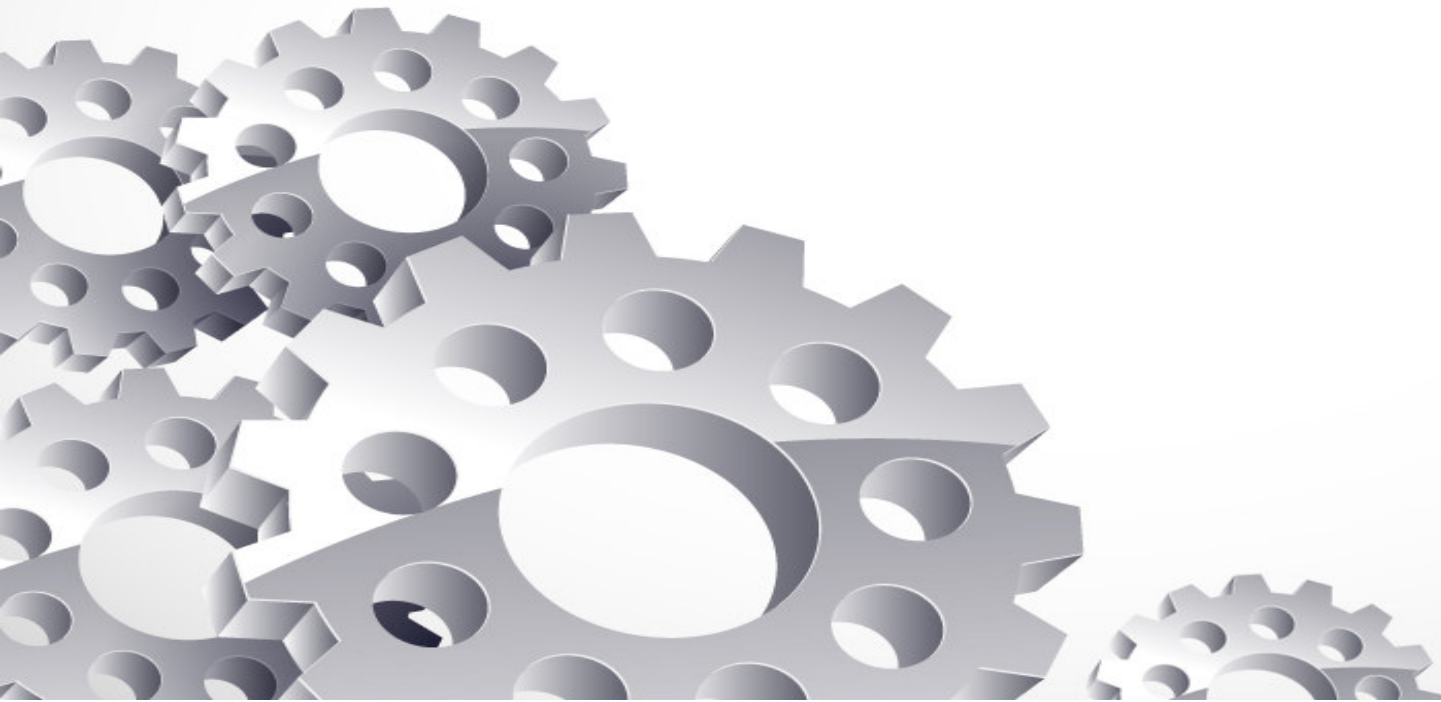
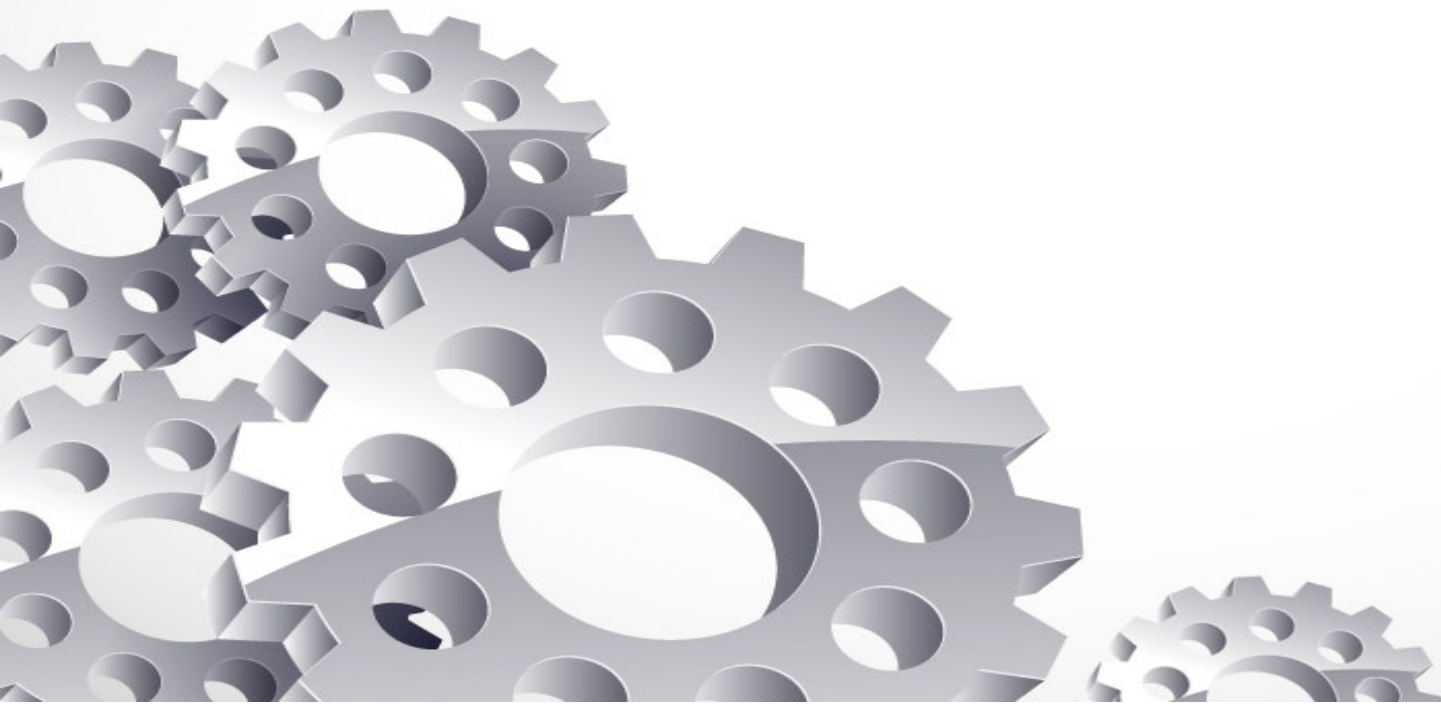


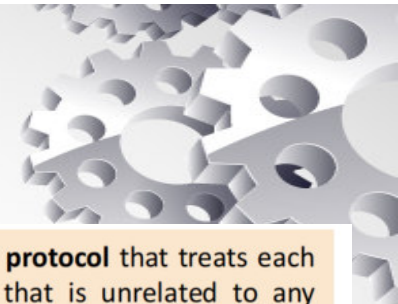
IoT Data Collection and Control Using MQTT, REST API & Cloud Services



Rest API & MQTT



HTTP (REST)



HTTP (REST)

Client – Server based

Stateless request/response protocol

- Client request information from server WebService (HTTP request)

Data representation in URL/URI form

e.g. <http://test.nyp.edu/docs/test.html>, format could be in json, xml etc.



- Server's WebService responds to the requests (HTTP Response)

A response is made up of a three-digit status code, some headers and some optional content.

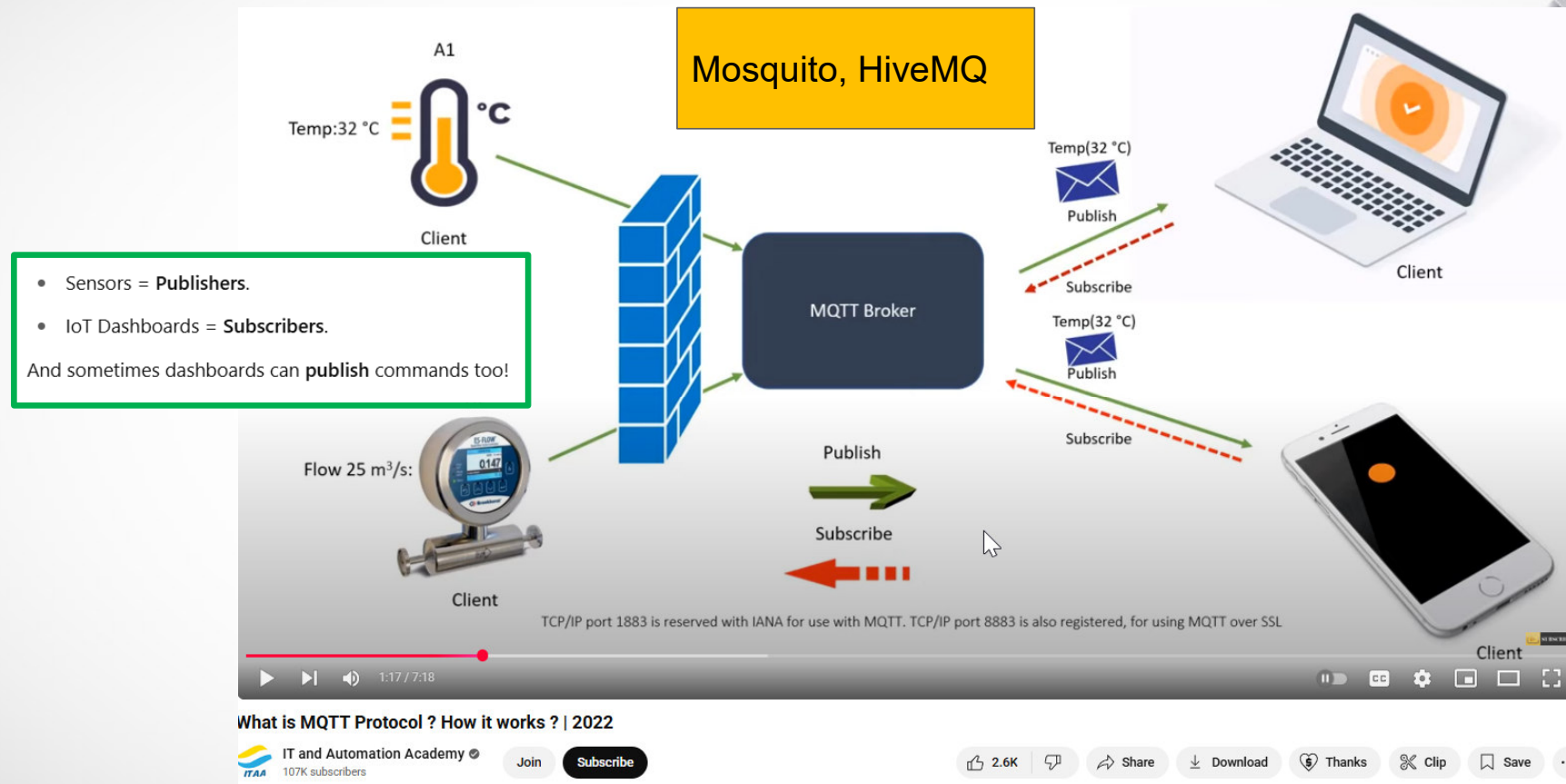


Rest API Functions and Analogy

A REST API is a way for two devices or programs to communicate with each other using the HTTP protocol (the same protocol your web browser uses).

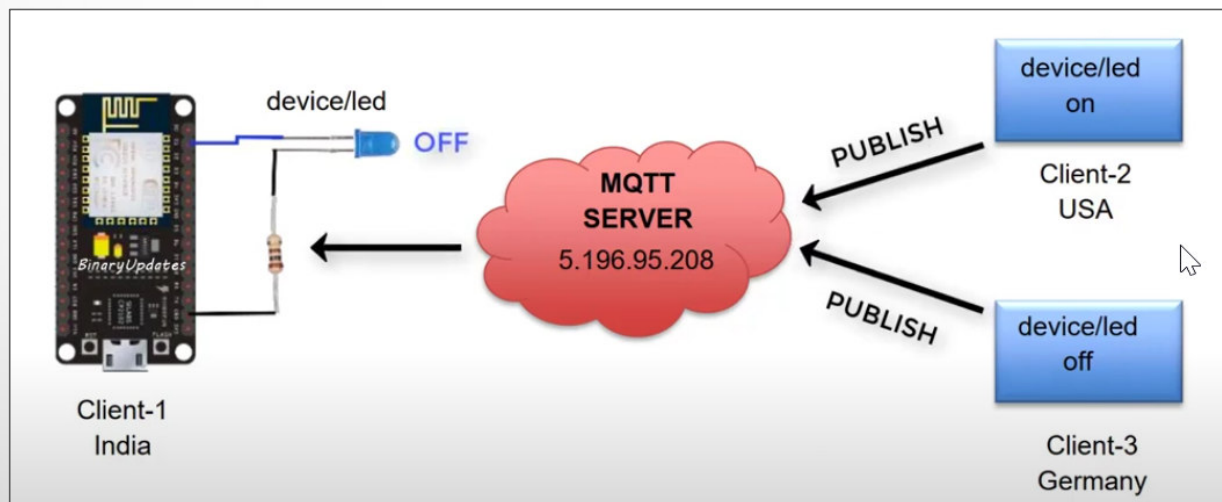
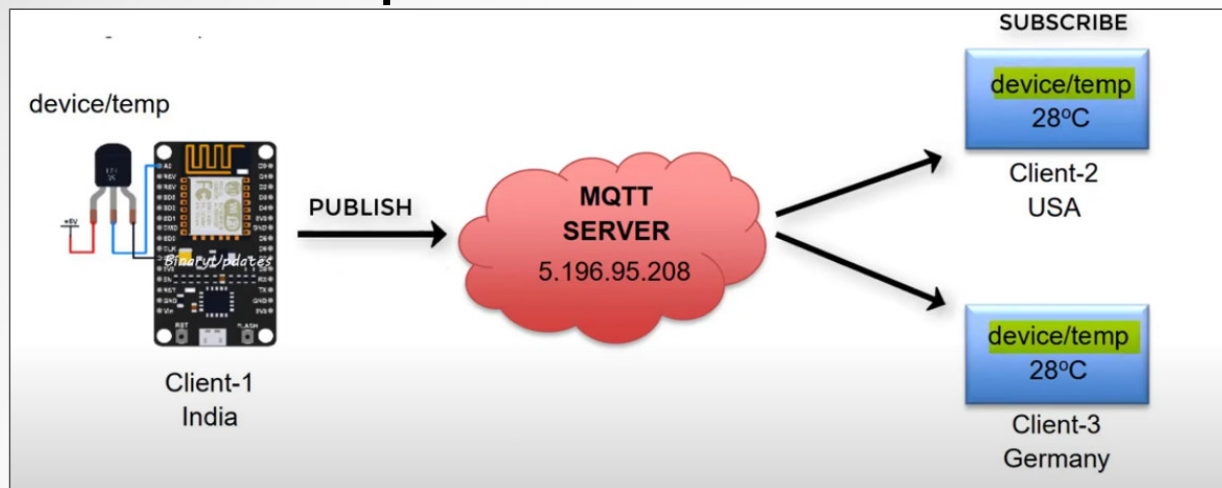
Function	HTTP Method	Analogy	Common Use
GET	GET	You ask: "Show me the menu." → You receive the menu.	Read data from server
POST	POST	You order a new dish → Kitchen prepares it → New dish added.	Send new data to server (create resource)
PUT	PUT	You update your order: "Change my pizza to extra cheese."	Update existing data
DELETE	DELETE	You say: "Cancel my order."	Delete data from server

MQTT = Message Queuing Telemetry Transport.



What is MQTT Protocol ? How it works ? | 2022

MQTT Uplink and Downlink



PUBLISH is mechanism used to **SEND** Messages

SUBSCRIBE is mechanism used to **RECEIVE** Messages

TOPIC is like address or name, e.g. for temp. sensor we assign topic as **device/temp**

PAYLOAD is an actual MESSAGE or **DATA to be SENT** from Client

MQTT QoS (Quality of Service) Levels



What is QoS?

- QoS defines the level of guarantee for message delivery between the sender and receiver in MQTT communication.
- Higher QoS levels ensure more reliable delivery but may introduce more overhead.

QoS Level	Name	Guarantee	Typical Use
0	At most once	Message is sent once , no acknowledgment. If it is lost, it is lost.	Sensor data that is not critical (e.g. temperature updates every 1s)
1	At least once	Message is sent one or more times until acknowledgment is received. Duplicates may occur.	Important data where duplicates are acceptable (e.g. control commands)
2	Exactly once	Message is sent exactly once with a two-phase handshake to ensure no duplication and no loss.	Critical data where duplication or loss is not acceptable (e.g. billing transactions, actuator controls)



REST API vs MQTT

REST API is best for:

One-time **requests** (device config, registration)

Management and control of IoT devices

HTTP-friendly → easy to integrate with web apps

MQTT is best for:

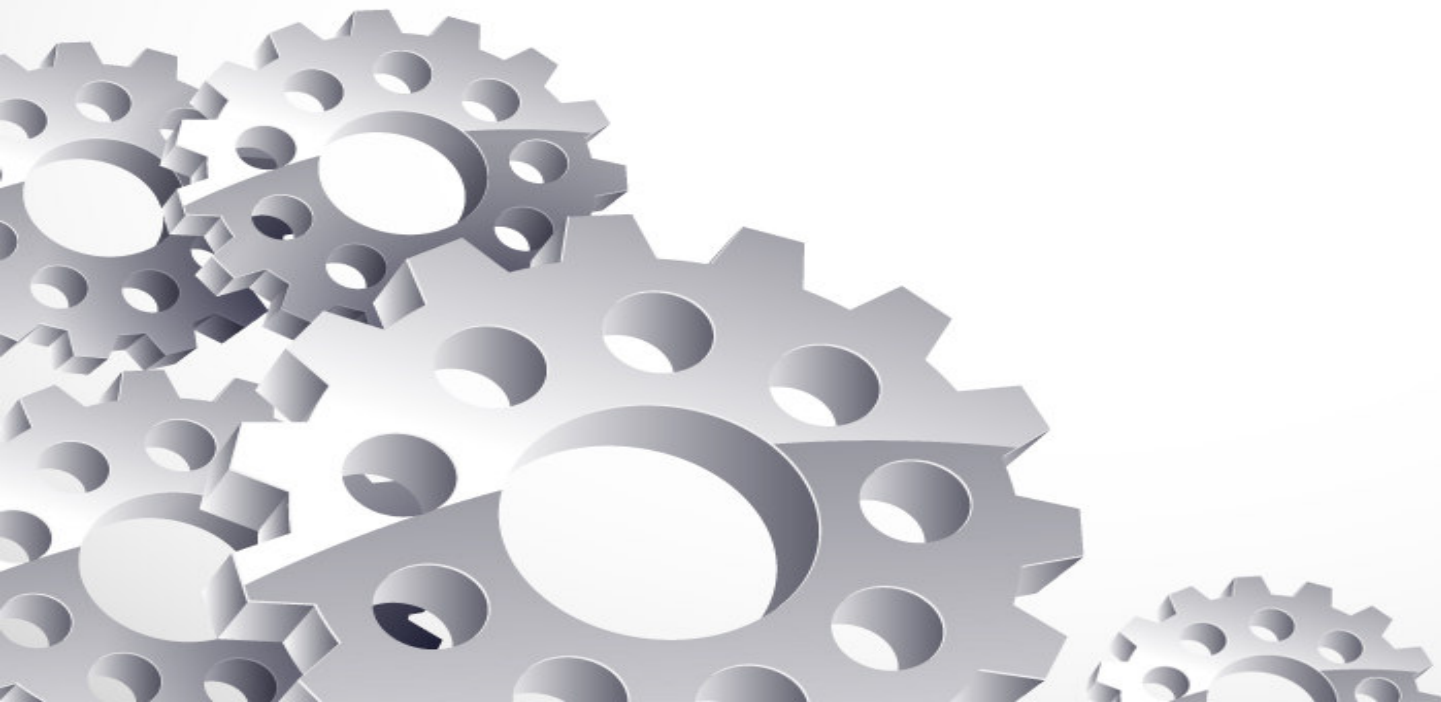
Real-time streaming of data

Low bandwidth or **low power** IoT networks

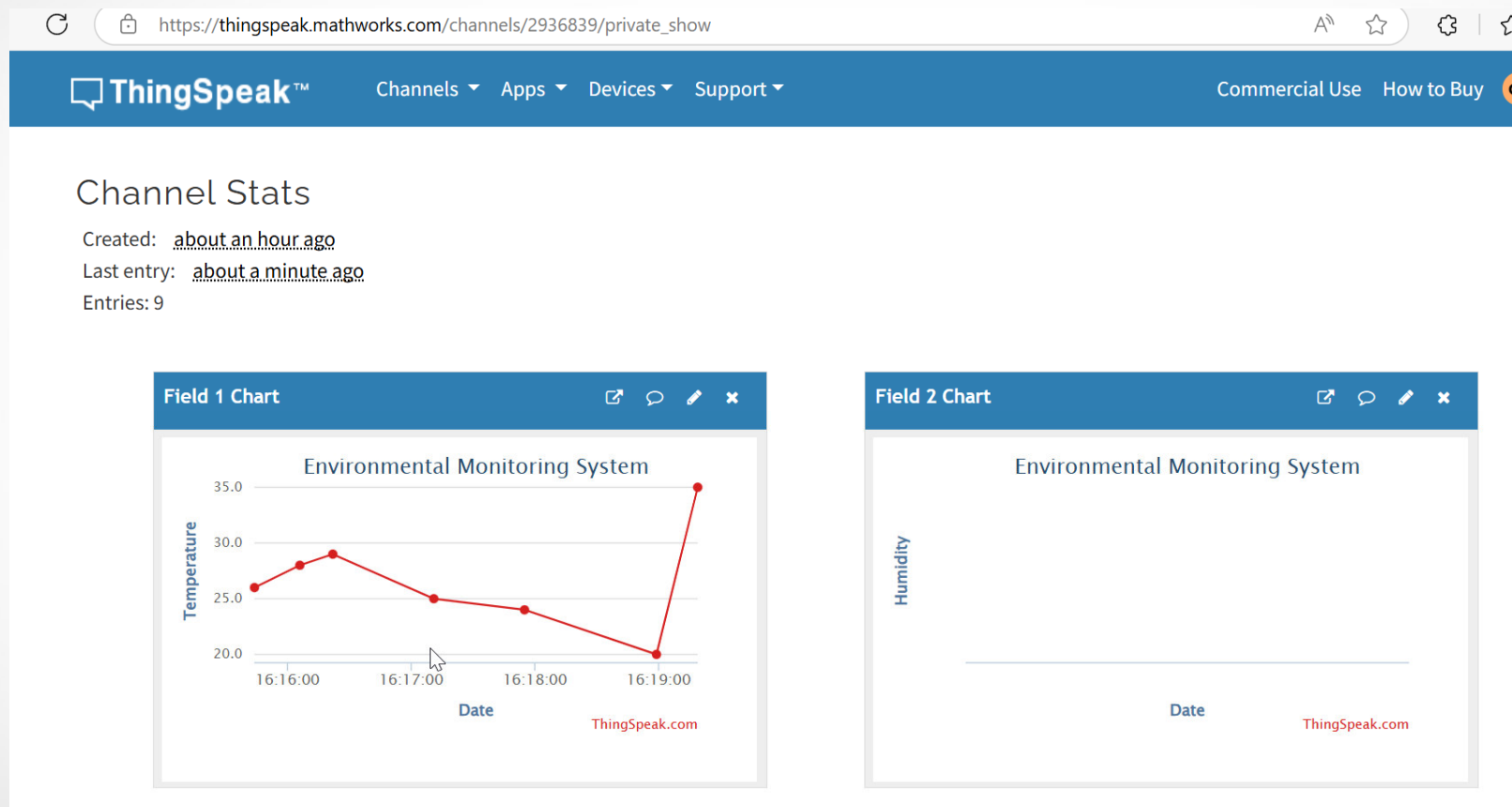
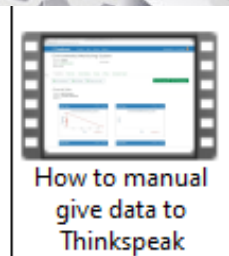
Push model → subscribers receive updates automatically

Feature	REST API	MQTT
Model	Request-Response	Publish-Subscribe
Protocol	HTTP	MQTT (lightweight over TCP/IP)
Connection	Short-lived (new connection per request)	Long-lived (persistent connection)
Data Format	JSON, XML, HTML	JSON, binary, plain text

IoT Dashboard - Thing Speak



ThingSpeak – from Matlab



Think Speak – REST API Setup

The screenshot shows the Thingspeak REST API setup page for an 'Environmental Monitoring System'. The browser address bar displays the URL: `https://api.thingspeak.com/update?api_key=Y968PL4FA9FZ4UFC&field1=35`. On the left, a box contains the number '10' with an arrow pointing to it from a grey box labeled 'Wait for 15s to refresh'. Below this, an 'Overall' section shows the URL components: `api.thingspeak.com/update?api_key=` followed by a grey box '+ API key', then `+ &field1=35` with an arrow pointing to a green box labeled 'payload'. On the right, the Thingspeak interface shows the 'Write API Key' section with a text input field containing 'Y968PL4FA9FZ4UFC' and a yellow button labeled 'Generate New Write API Key'. A line connects this key to the 'api_key=' part of the URL in the address bar. The Thingspeak header includes navigation links: Channels, Apps, Devices, and Support. The page title is 'Environmental Monitoring System' with details: Channel ID: 2936839, Author: mwa0000037664088, and Access: Private. Tabs for 'Private View', 'Public View', 'Channel Settings', 'Sharing', and 'API Keys' are visible.

10

Wait for 15s to refresh

Overall

api.thingspeak.com/update?api_key=

+ API key

+ &field1=35

payload

Thingspeak™ Channels Apps Devices Support

Environmental Monitoring System

Channel ID: 2936839
Author: mwa0000037664088
Access: Private

Sensor Data

Private View Public View Channel Settings Sharing API Keys

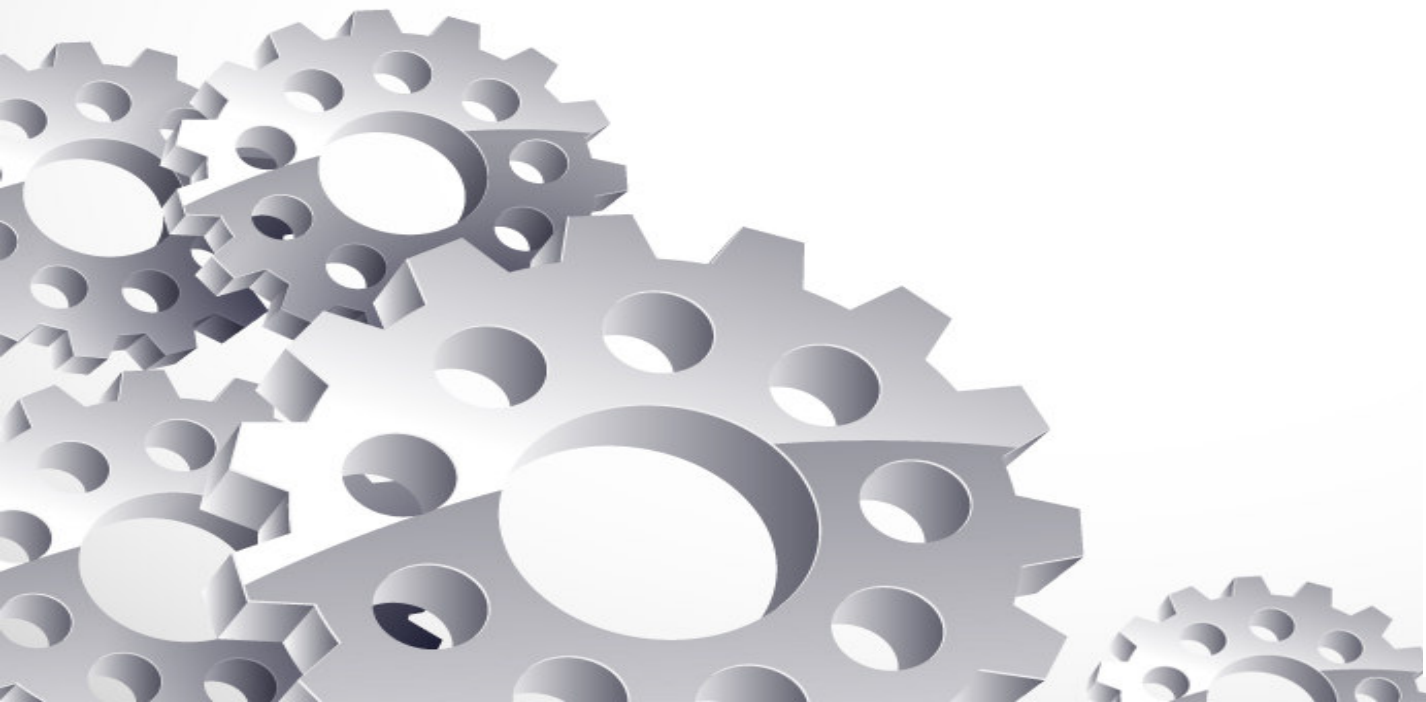
Write API Key

Key Y968PL4FA9FZ4UFC

Generate New Write API Key

- API Key = Your Device's ID Card.
- Without it: No Access.
- It's a small security price for safe, private, and controlled data.

IoT Dashboard - Adafruit IO



Get Started

FREE

forever

30 data points per minute
30 days of data storage
Actions every **15 minutes**
5 dashboard limit
2 WipperSnapper device limit
5 group limit
10 feed limit

Sign Up Now

Official (Open)

Adafruit IO – MQTT Setup

Downlink Control Example

```
Thonny - MicroPython device : /Adafruit_MQTT_PubAndSub.py 32:9
File Edit View Run Tools Help
[ Adafruit_MQTT_PubAndSub.py ] - [ credential.py ]
10 import credential
11 import gc
12 gc.collect()
13
14 sensor = dht.DHT22(Pin(15)) # read temperature
15 led = Pin(2,Pin.OUT) # turn on LED with switch
16 #print(led)
17 #print(sensor)
18
19 # the following function is the callback which is called when subscribed data
20
21 def cb(topic, msg):
22     print('Subscribe: Received Data: Topic = {}, Msg = {}'.format(topic, msg))
23     try:
24         message = msg.decode('utf-8').strip().lower()
25
26         if message in ['1', 'on', 'true']:
27             led.value(1)
28         else:
29             led.value(0)
30     except Exception as e:
31         print("Error in callback:", e)

```

```
Shell
MPY: soft reboot
Connected to WiFi. IP: 192.168.50.88
after create random MQTT Client ID
after ADAFRUIT Setting
after Client setting
after client connect 1
after client connect 2
after feed setting
after accum time setting

```



Adafruit
downlink control
LED - camera
mode



Adafruit MQTT
downlink control
LED

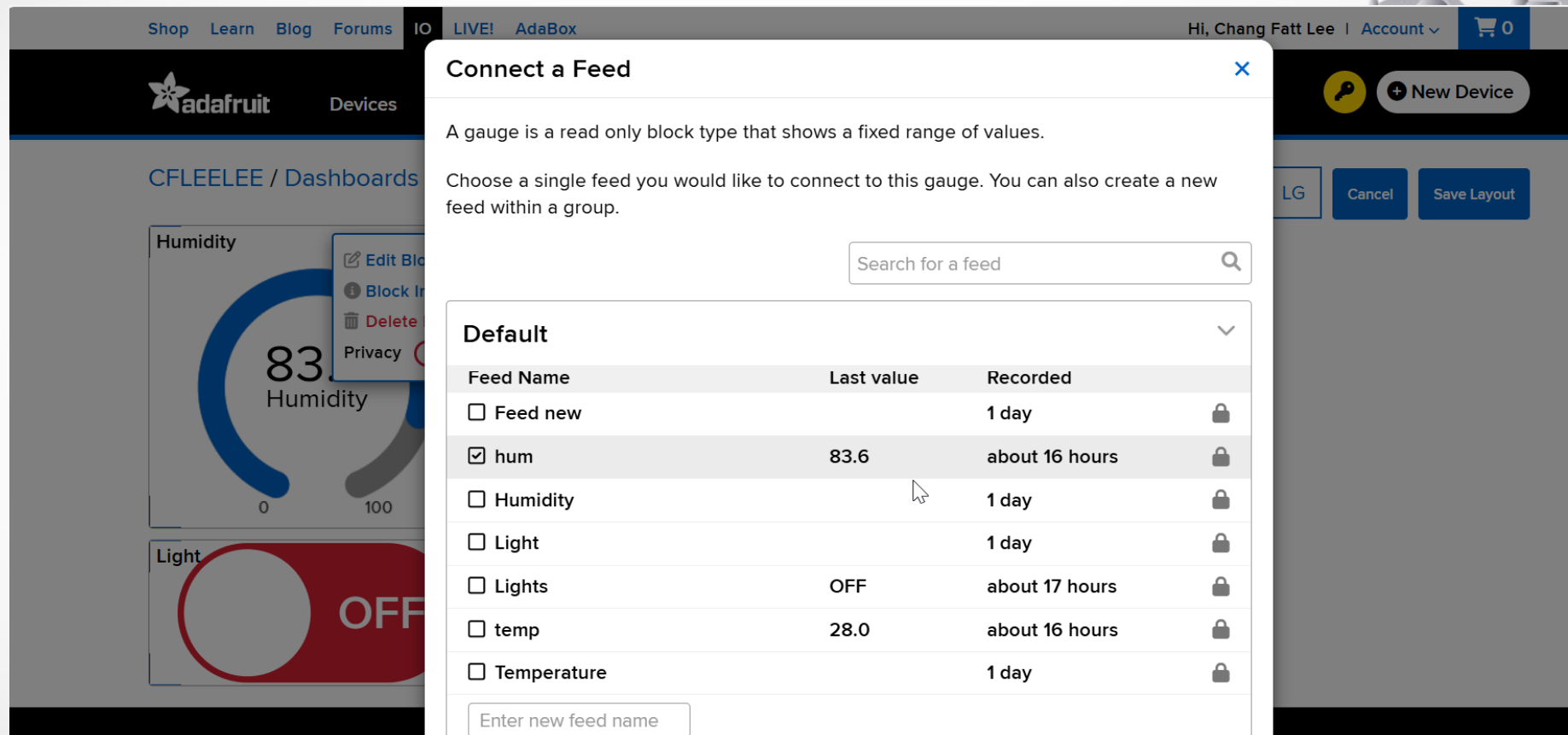
Official (Open)

Create New Block



The screenshot displays the Adafruit IO web interface. The browser address bar shows the URL `https://io.adafruit.com/CFLEELEE/dashboards/home-automation`. The top navigation bar includes links for Shop, Learn, Blog, Forums, IO, LIVE!, and AdaBox. The main header features the Adafruit logo and navigation tabs: Devices, Feeds, Dashboards, Actions, and Power-Ups. A user profile icon and a 'New Device' button are on the right. The breadcrumb trail reads 'CFLEELEE / Dashboards / Home Automation'. A 'Dashboard Settings' panel is open on the right, containing options: Edit Layout, Create New Block (highlighted with a mouse cursor), View Fullscreen, Dark Mode (on), Block Borders (on), Dashboard Privacy (off), and Delete Dashboard. The main content area shows 'IO Status' as online, a quote by Dean Kamen, and a list of links for help and documentation. The footer includes the Adafruit logo, social media icons, and a statement about being a Minority and Woman-owned Business Enterprise (M/WBE).

Feed



Connect a Feed

A gauge is a read only block type that shows a fixed range of values.

Choose a single feed you would like to connect to this gauge. You can also create a new feed within a group.

Search for a feed

Feed Name	Last value	Recorded	
☐ Feed new		1 day	
☒ hum	83.6	about 16 hours	
☐ Humidity		1 day	
☐ Light		1 day	
☐ Lights	OFF	about 17 hours	
☐ temp	28.0	about 16 hours	
☐ Temperature		1 day	

Enter new feed name

A feed in Adafruit IO is where your IoT device **sends (writes)** or **receives (reads)** a value. It's a basic building block for dashboards and device automation.

Official (Open)

Dashboard



Devices

Feeds

Dashboards

Actions

Power-Ups



+ New Device

CFLEEELEE / Dashboards / Nightclass Dashboard



Reset

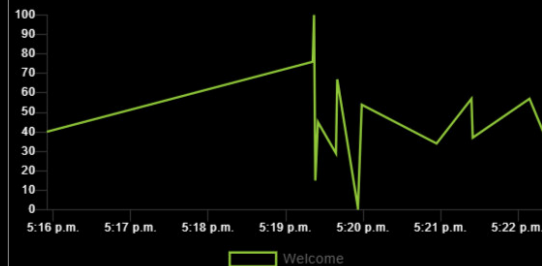
Try Me!

17

OFF

17
Value

May 9th 2025, 5:22:39PM



What are the three steps in the create block form?

1. Select block type
2. Select feed(s)
3. Block settings



Control m5 led
from Adafruit

Thank you

